

Apurva Aggarwal

201-551-9782 | aggarw86@msu.edu | [linkedin.com/in/apurva0510](https://www.linkedin.com/in/apurva0510) | github.com/apurva0510

EDUCATION

Michigan State University

Bachelor of Science in Computer Science, Minor in Business

East Lansing, MI

May 2027

- **GPA: 3.95 — Dean's List, 5 semesters**
- **Relevant Coursework:** Data Structures & Algorithms, Artificial Intelligence, Big Data Analysis, Cloud Development

TECHNICAL SKILLS

Languages: Python, SQL, C/C++, JavaScript, R, ARM64 Assembly

Data/ML: Pandas, NumPy, scikit-learn, PyTorch, TensorFlow, PostgreSQL, SQLAlchemy

Cloud/Systems: Azure, Docker, Git, GitHub Actions, REST APIs, Flask, Docker Compose, CI/CD

Analytics: Power BI, Tableau, Excel, KPI Design, A/B Testing

EXPERIENCE

Data Analytics Engineering Intern – Tesla, Inc., Residential Energy | Palo Alto, CA

Jan 2026 – Aug 2026

- Engineered a production **entity resolution pipeline** for a **35K-account installer network**, combining fuzzy matching and ranking techniques to achieve **90%+ accuracy** while reducing reconciliation effort by **280 hours per 10K records**
- Built internal **GenAI workflow agents** supporting **100+ sales and operations users** across **20 workflows**, combining enterprise knowledge retrieval and Jira, Teams, Confluence, and mail integrations to automate business processes
- Owned executive-facing **solar product analytics**, defining KPI methodologies and tracking orders, deliveries, goal attainment, and rebate adoption across weekly, monthly, and quarterly business reviews
- Developed **performance benchmarking frameworks** across **4K Certified Installers**, leveraging CRM, delivery, commission, and growth metrics to inform account prioritization and sales performance management

Technology Consulting Intern – Ernst & Young (EY) | Singapore

May 2025 – Aug 2025

- Led client-facing **UAT** for a \$13M enterprise migration, validating **10K+ records** and surfacing critical defects before release
- Automated migration validation workflows using **SQL Server**, reducing manual QA effort by **50%**
- Supported migration of legacy file storage to **Azure Blob Storage**, improving access control and cloud readiness
- Triaged defects in **Azure DevOps**, distinguishing missing vs incorrect data issues to accelerate sprint resolution

Data Analytics Fellow – The Global Career Accelerator | Remote

Aug 2024 – Dec 2024

- Analyzed **30M+ engagement records** for a **Recording Academy-sponsored** audience strategy project, uncovering **5x higher engagement rates** for photo-based content
- Evaluated Grammy.com site performance using **pages/session, bounce rate, time on site, and mobile traffic KPIs**, recommending continued site separation
- Designed **A/B testing analyses** with hypothesis testing and significance testing to validate differences in engagement metrics at 95% confidence
- Modeled regional **energy availability and renewable capacity** for an **Intel-sponsored** sustainability project, ranking U.S. regions for data center readiness

PROJECTS

Argus | Python, Streamlit, PostgreSQL, SQLAlchemy, GitHub Actions

Jun 2026

- Built a **market intelligence platform** tracking 55 AI, semiconductor, and quantum companies using Streamlit, PostgreSQL, and SQLAlchemy
- Architected a **27-table PostgreSQL warehouse** and scheduled ETL pipelines with GitHub Actions to ingest market, filing, earnings, news, and macroeconomic data
- Designed scoring models, thematic indices, and watchlist alerts to surface pullback opportunities and sector-wide trends

FarmX (MHacks 2024 Winner) | Python, scikit-learn, Streamlit

Sept 2024

- Built an ML application using **Random Forest models** to recommend crop suitability and NPK nutrient levels from soil and weather inputs
- Engineered preprocessing and feature pipelines with **Pandas, NumPy, and scikit-learn**, deploying a Streamlit prototype
- Won **Best Sustainability Track** at MHacks 2024 for applying ML to optimize farming input recommendations

Micro Foods Market (Backend) | Docker, Flask, REST APIs, SQLite, JWT

Apr 2025

- Designed a **microservices backend** with 5 Flask services for authentication, products, search, orders, and logging
- Implemented **JWT authentication**, REST APIs, SQLite persistence, and Docker Compose networking for local orchestration
- Structured service-to-service workflows for orders, product search, and logging to separate business logic across backend services

LEADERSHIP & INVOLVEMENT

SpartaHack – Co-Director for 600+ attendee student-run hackathon

International Students Association – Director of Technical Affairs, leading development of centralized affiliate web platform

Resident Assistant – Mentored and supported 80+ students in residential community